

## Practice Problems

### Problem 1.1

You invest \$3,200 in a savings account on January 1, 2004. On December 31, 2004, the account has accumulated to \$3,294.08. What is the annual interest rate?

### Solution

Let  $r$  be the annual interest rate. Since the investment grows for one year,

$$3200(1 + r) = 3294.08.$$

Solving for  $r$ ,

$$1 + r = \frac{3294.08}{3200}$$

and therefore

$$r = \frac{3294.08}{3200} - 1.$$

Now,

$$r = 1.0294 - 1 = 0.0294.$$

Hence, the annual interest rate is

$$\boxed{2.94\%}.$$